Improving Equilibrium Propagation Without Weight Symmetry Through Jacobian Homeostasis

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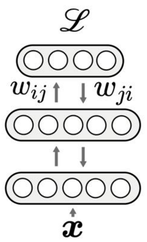
Generalized Equilibrium Propagation [1]

Goal: Local gradients for Neurobiology and Neuromorphics



Dynamics: $\frac{\mathrm{d}\boldsymbol{u}}{\mathrm{d}t} = F(\boldsymbol{\theta}, \boldsymbol{u}, \boldsymbol{x}) + \beta \frac{\mathrm{d}\mathscr{L}}{\mathrm{d}\boldsymbol{u}}^\top$

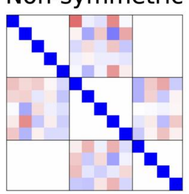
Fixed-point (prediction): $0 = F(\boldsymbol{\theta}, \boldsymbol{u}_0^*, \boldsymbol{x})$

EP error vector:

$$\left.\frac{\mathrm{d}\boldsymbol{u}^*}{\mathrm{d}\beta}\right|_{\beta=0} = -J_F(\boldsymbol{u}_0^*)^{-1} \cdot \frac{\partial F}{\partial \beta}(\boldsymbol{u}_0^*)$$

$$= -\underbrace{J_F(\boldsymbol{u}_0^*)^{-\top}}_{\text{for EBMs}} \cdot \underbrace{\frac{\mathrm{d}\mathcal{L}}{\mathrm{d}\boldsymbol{u}_0^*}^\top}_{\text{by Def. 1}} = \boldsymbol{\delta}$$

Non-symmetric



Entangled sources of bias:
- Finite differences ❌
- Jacobian asymmetry ❌

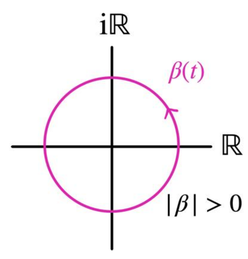
Our contributions:

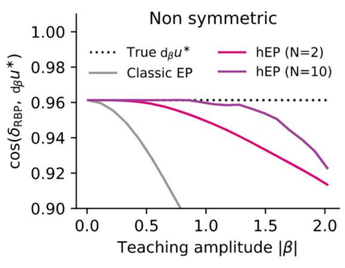
- Disentangle the two sources of bias in non-symmetric EP
- Extend holomorphic EP [2] to non symmetric networks
- New homeostatic loss promotes Jacobian symmetry

Accurate computation of the EP error vector

For complex-differentiable F:

$$\left.\frac{\mathrm{d}\boldsymbol{u}^*}{\mathrm{d}\beta}\right|_{\beta=0} = \frac{1}{T|\beta|}\int_0^T \boldsymbol{u}^*_{\beta(t)} e^{-2\mathrm{i}\pi t/T}\mathrm{d}t.$$





Derivative as integral ✅
Non vanishing perturbations ✅

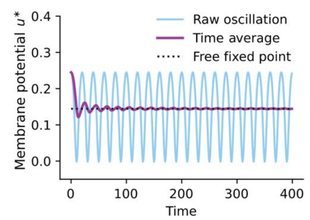
Continuous-in-time estimation of the gradient

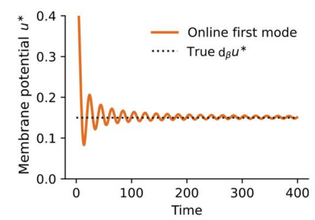
Presynaptic term
(Mean value theorem):

$$\frac{1}{t}\int_0^t \frac{\partial F}{\partial \boldsymbol{\theta}}(\boldsymbol{u}^*_{\beta(\tau)})\mathrm{d}\tau \xrightarrow[t\to\infty]{} \frac{\partial F}{\partial \boldsymbol{\theta}}(\boldsymbol{u}_0^*)$$

Post synaptic error:

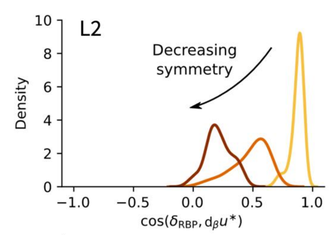
$$\frac{1}{t|\beta|}\int_0^t \boldsymbol{u}^*_{\beta(\tau)} e^{-2\mathrm{i}\pi\tau/T}\mathrm{d}\tau \xrightarrow[t\to\infty]{} \frac{\mathrm{d}\boldsymbol{u}^*}{\mathrm{d}\beta}$$

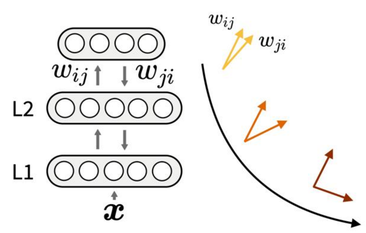


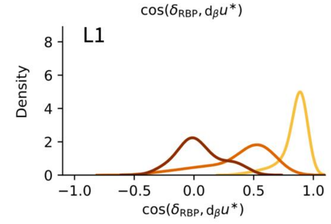


The gradient estimate can be computed in continuous time without discrete forward/backward phases ✅

Isolated effect of Jacobian asymmetry







General transformation:

$$\mathrm{d}_\beta \boldsymbol{u}^* = J_F(\boldsymbol{u}_0^*)^{-1} J_F(\boldsymbol{u}_0^*)^\top \boldsymbol{\delta}$$

At first order:

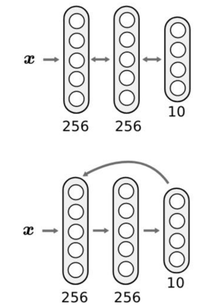
$$\mathrm{d}_\beta \boldsymbol{u}^* = \boldsymbol{\delta} - 2S^{-1}A\boldsymbol{\delta} + o(S^{-1}A\boldsymbol{\delta})$$

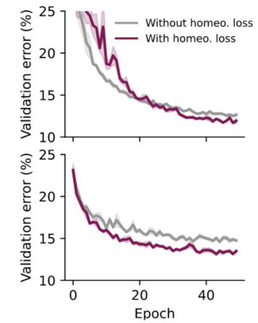
New homeostatic loss:

$$\mathcal{L}_{\text{homeo}} := \mathop{\mathbb{E}}_{\boldsymbol{\varepsilon}\sim\mathcal{N}(0,I)} \left[\underbrace{\|J_F\boldsymbol{\varepsilon}\|^2}_{\text{Minimize sensitivity to perturbation}} - \underbrace{\boldsymbol{\varepsilon}^\top J_F^2 \boldsymbol{\varepsilon}}_{\text{Increase second-order reconstruction}}\right]$$

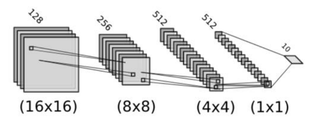
Benefits of the homeostatic loss on learning

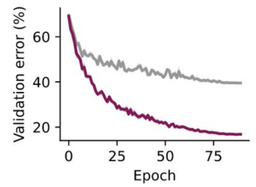
Fashion MNIST





CIFAR-10





	CIFAR-10	ImageNet 32×32	
	Top-1 (%)	Top-1 (%)	Top-5 (%)
hEP w/o $\mathcal{L}_{\text{homeo}}$	60.4 ± 0.4	–	–
$\text{hEP}_{N=2,\lvert\beta\rvert=1}$	81.4 ± 0.1	–	–
hEP (True $\mathrm{d}_\beta \boldsymbol{u}^*$)	84.3 ± 0.1	31.4 ± 0.1	55.2 ± 0.1
RBP	87.8 ± 0.3	–	–
RBP w/o $\mathcal{L}_{\text{homeo}}$	87.7 ± 0.2	–	–
Sym. $\text{hEP}_{N=2,\lvert\beta\rvert=1}$	88.6 ± 0.2	36.5 ± 0.3	60.8 ± 0.4

Homeostatic loss is:
- More general than symmetrising weights ✅
- Reduces the gap with the energy-based case ✅

References

[1] Scellier, et al Generalization of equilibrium propagation to vector field dynamics. arXiv:1808.04873, 2018.
[2] Laborieux and Zenke. Holomorphic equilibrium propagation computes exact gradients through finite size oscillations. NeurIPS 2022.

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